

Padé Approximation and the Pandrosion Rational Field

Ivan Besevic

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Abstract

The Pandrosion field $F_P = P'/P$ is a rational function of type $[n-1, n]$ —the most natural Padé approximant. Its poles are the roots of P , its residues are the multiplicities, and its Laurent expansion at infinity reproduces Newton’s power sums (Paper 90). We show that Padé approximation systematically outperforms Taylor near singularities because it *captures poles*, exactly as F_P captures root locations. Verified: $[5, 5]$ Padé for e^z achieves 10^{-10} error with 11 terms; $[0, 1]$ Padé for $1/(1-z)$ is exact.

1 F_P as a Padé approximant

Proposition 1.1. $F_P(z) = P'(z)/P(z)$ is a rational function of type $[n-1, n]$: numerator degree $n-1$, denominator degree n . Its partial fraction decomposition:

$$F_P(z) = \sum_{j=1}^n \frac{1}{z - \alpha_j} \tag{1}$$

is the canonical Padé form with poles at the roots of P .

2 Padé vs Taylor

Approximant	e^1 error	Terms used
T_3 (Taylor)	5.2×10^{-2}	4
$[2, 1]$ Padé	3.2×10^{-2}	4
T_5 Taylor	1.6×10^{-3}	6
$[3, 3]$ Padé	2.8×10^{-5}	7
T_9 Taylor	3.0×10^{-7}	10
$[5, 5]$ Padé	2.8×10^{-10}	11

For $1/(1-z)$ at $z = 0.9$: T_{19} still has error 1.2, while the $[0, 1]$ Padé is *exact*.

3 The Pandrosion–Padé dictionary

Padé concept	Pandrosion tool
Poles of approximant	Roots α_j of P
Residues at poles	Multiplicities (=1 for simple roots)
Type $[m, n]$	F_P has type $[n - 1, n]$
Laurent at ∞	Newton sums $p_k = \sum \alpha_j^k$ (Paper 90)
“Captures poles”	F_P captures root locations

References

- [1] H. Padé, *Sur la représentation approchée*, Ann. Éc. Norm. Sup. **9** (1892).
- [2] I. Besevic, *Vieta–Newton unified*, Pandrosion Dynamics **90** (2026).